Lecture 27.04: Exploratory Data Analysis (EDA) — Distribution, Skewness & Outliers

📌 Executive Summary

Rigorous exploratory data analysis on the CreditWise dataset, target class imbalance visualization, univariate income distribution skewness, and Tukey boxplot outlier diagnostics.

🏛️ Theoretical Formulation & Mathematical Architecture

📊 Exploratory Statistical Analysis Protocol

Exploratory Data Analysis (EDA) serves as the diagnostic lens revealing underlying probability densities, feature correlations, and structural pathologies before statistical model training.

1. Target Class Imbalance Quantification:

The binary response variable $Y \in \{0, 1\}$ exhibits an empirical prior distribution:

> P(Y = 0) = rac{N_0}{N} pprox 68.6\% \qquad ext{and}\qquad P(Y = 1) = rac{N_1}{N} pprox 31.4\%

This baseline imbalance implies that a naive zero-rule classifier predicting 'Rejected' for every single applicant achieves an apparent accuracy of 68.6%, while completely failing to execute any commercial banking operations.

📦 Tukey's Five-Number Summary & Outlier Fences

To mathematically inspect extreme leverage points and potential data recording errors, we apply John Tukey's Interquartile Range (IQR) fence rule:

Given the ordered rank distribution $\mathcal{D} = \{x_{(1)}, x_{(2)}, \dots, x_{(n)}\}$:

> $$extIQR = Q_3 - Q_1$$

The lower and upper outer fences are established at:

$$F_{extLower} = Q_1 - 1.5imesextIQR \qquad extand \qquad F_{extUpper} = Q_3 + 1.5imesextIQ$$

Any observation x_i outside $[F_{extLower}, F_{extUpper}]$ is designated an empirical outlier. In financial underwriting:

- Outliers in `Applicant_Income` or `Savings` represent legitimate high-net-worth individuals (HNWIs) rather than measurement errors.
- Outliers in `DTI_Ratio` (> 1.0) signify hyper-leveraged borrowers who spend more on existing debt than their total monthly income.

```
flowchart TD
    A["Raw Feature Values"] --> B["Calculate Q1 (25th) & Q3 (75th Percentiles)"]
    B --> C["Compute IQR = Q3 - Q1"]
    C --> D["Establish Fences: [Q1 - 1.5*IQR, Q3 + 1.5*IQR]"]
    D --> E{"Point Within Fences?"}
    E -->|"Yes"| F["Typical Underwriting Sample"]
    E -->|"No"| G["Financial Leverage Outlier (HNWI or Severe Debt)"]
```

💻 Vectorized Python Implementation

```
import pandas as pd
import numpy as np

# Load preprocessed dataframe
df = pd.read_csv("loan_approval_data.csv")

# Compute class balance
class_counts = df["Loan_Approved"].value_counts(normalize=True)
print("Target Distribution:")
print(class_counts)

# Outlier bounds for Credit_Score and DTI_Ratio
for col in ["Credit_Score", "DTI_Ratio", "Applicant_Income"]:
    series = df[col].dropna()
    q1 = series.quantile(0.25)
    q3 = series.quantile(0.75)
    iqr = q3 - q1
    lower_fence = q1 - 1.5 * iqr
    upper_fence = q3 + 1.5 * iqr
    outliers = series[(series < lower_fence) | (series > upper_fence)]
```

```
    print(f"{col} Outliers count: {len(outliers)} (Bounds: [{lower_fence:.2f}, {up
```

🔬 Key Engineering Takeaways & Invariants

1. **Mathematical Grounding**: Every transformation and classification step directly addresses the underlying empirical distribution and financial risk constraints.
2. **Reproducibility**: Clean, leak-free pipeline partitioning guarantees that performance metrics match production operating realities.
3. **Domain Alignment**: Aligning model evaluation with the asymmetric cost of financial errors protects institutional solvency.